

Gene Technology - Concerns

1. Herbicide resistant crops may be produced by genetic engineering.

(c) Suggest **one** advantage to a farmer of growing herbicide-resistant crops.

(1 mark)

(d) Many people are opposed to the growing of herbicide-resistant crops produced in this way.

Suggest **one** reason why.

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.....

(1 mark)

2.

Use the above information and your own knowledge and understanding to evaluate how the production of rabbit–human embryos may help research into human diseases.

Remember to give a conclusion as part of your evaluation.

(4 marks)

3.

Read the passage about cloning cattle.

The Government has been accused of 'inexcusable behaviour' because a calf of a cloned American 'champion' cow has been born on a British farm. Campaigners say it will undermine trust in British food because the cloned cow's milk could enter the human food chain.

But supporters of cloning say that milk from clones and their offspring is as safe as the milk we drink every day.

Those in favour of cloning say that an animal clone is a genetic copy. It is not the same as a genetically engineered animal. Opponents of cloning say that consumers will be uneasy about drinking milk from cloned animals.

Use the information in the passage and your own knowledge and understanding to evaluate whether the government should allow the production of milk from cloned 'champion' cows.

Remember to give a conclusion to your evaluation.

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(5 marks)

4.

Some people might object to the growing of genetically-engineered, high-sugar rye grass for feeding cattle.

Give **two** reasons why.

1

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2

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(2 marks)

5.

People could ask scientists to use this technique to clone long-dead relatives, whose bodies have been deep-frozen.

Most people would be opposed to cloning a human from a deep-frozen, long-dead relative.

Give **one** reason why.

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(1 mark)

3.

pros eg

any two from:

- overcomes shortage of human eggs / rabbits produce lots of eggs
- ethical / religious issues with using human embryos
- reduces tests on (adult) humans
- may provide cure for / cause of disease
- embryo not allowed to develop beyond 14 days
- no harm to rabbit
- 99.5 % human genetic information so very similar to human or will react in the same way

cons eg

any two from:

- ethical / religious objections to mixture of human and rabbit genes
- ethical issues with experimenting with rabbits
- ethical / religious objections to killing embryos
- 0.5% of rabbit genetic information might affect results
- 14 days too short a time to get results

to obtain 3 marks candidates must give **one** reasonable pro **and one** reasonable con

ignore all embryos identical

allow some people object to using rabbits / cruel to rabbits

max 2

max 2

4.

any two from eg

- concern about effect on (health) of cow
- concern about effects on human (health)
- concern about food chain effects or effects on ecosystem
- effect on gene pool

ignore not natural or cost

ignore ethical / religious arguments

if no other marks awarded

'we don't know the long term effects' = 1 mark

5.

ethical / religious / emotional reasons

or

not known if it is safe / long term effects not known

ignore playing God / unnatural / immoral

1